

PRACTICAL 5

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Design Rationale

CampusGuard is centred around an Incident, which stores the incident's descriptive information, reports and operational status. The incident owns its IncidentStatus, allowing its lifecycle to be modelled through the State pattern as REPORTED, ACTIVE, RESOLVED and CANCELLED. The incident does not own its CampusComponent location because campus components form part of an independently managed campus hierarchy.

A Report represents information about a change or required response during an incident. It contains the relevant incident, affected campus area, report type and optional response information. Reports are passed to the Coordinator, which acts as the Mediator to Service Colleagues. This allows a response from one part of the system to result in coordinated actions by other services without creating many-to-many dependencies.

Medical, Security and Facilities services are response services that manage ResponseUnit objects. A response unit is associated with its current incident, location and the response service that owns it, and it owns its UnitStatus. The State pattern is used to model the unit lifecycle, including AVAILABLE, DISPATCHED, OPERATING and UNAVAILABLE. The unit's incident reference is non-owning because the incident's lifetime is managed separately.

The campus is represented as a Composite structure consisting of CampusZone composites and CampusUnit leaves. Both can be treated uniformly as CampusComponent objects. Access-control operations can therefore be performed on any component in the hierarchy. Restricting or restoring access on a parent component recursively affects its children.

The Access Control subsystem demonstrates the Adapter pattern because the legacy access-control system exposes a different interface from the one required by CampusGuard. The legacy system stores the campus and locates components using an integer code, while CampusGuard operates directly on CampusComponent objects. The AccessControlAdapter translates between these interfaces, allowing the rest of CampusGuard to use the desired interface without depending on the legacy implementation.

Operator actions are represented using the Command pattern. RestrictAccessCommand, RestoreAccessCommand, SendAlertCommand, DispatchUnitCommand and HandleIncidentCommand encapsulate specific requests and invoke their appropriate receivers. The OperatorConsole acts as the Invoker, while the relevant services or Coordinator act as receivers. This separates the creation and invocation of requests from the objects that perform the actual operations.

The EmergencyDesk provides the Façade through which high-level emergency workflows can be performed without requiring the client to coordinate the individual services and commands itself. It provides three main workflows. respondToFire performs the predefined fire-response sequence automatically. handleIncident processes the reports already associated with an incident and coordinates the required responses. manageIncident provides ongoing management of an incident, allowing its current reports to be handled while leaving the incident active for further responses or reports. resolveIncident is used separately to complete the incident.

The Façade does not own the coordinator, services, console, incident or dispatched units that it references. These objects have lifetimes managed by the application or their respective owning objects. The console owns its commands, services own their response units, incidents and response units own their current status, and the campus hierarchy owns its child components. Non-owning pointers are therefore used where an object only references an object whose lifetime is managed elsewhere.

The patterns are integrated; the Façade workflow uses Commands to perform operations, the Coordinator Mediator to coordinate services when reports require additional responses, the Adapter when access control interacts with the legacy system, and the State and Composite patterns to manage incident/unit lifecycles and the campus hierarchy respectively.

Justification for State and Composite

Both Incident and ResponseUnit have status-dependent legal transitions. You can't resolve an incident that hasn't been activated, can't dispatch a unit that's already Operating. Without State, this becomes an if/else or switch on a status field scattered across every method that touches status.

The campus is a real tree, and access-control operations need to work the same way whether the target is one room or an entire zone; restricting a building should cascade to every room inside it.

GoF Participants

Command

Command	Command
ConcreteCommand	HandleIncidentCommand, DispatchUnitCommand, RestrictAccessCommand, RestoreAccessCommand, SendAlertCommand
Invoker	OperatorConsole
Receiver	Coordinator, ResponseService, AccessControlService, AlertService
Client	EmergencyDesk, main()

Mediator

Mediator	Coordinator
ConcreteMediator	CampusCoordinator
Colleague	Service
Concrete Colleague	SecurityService, MedicalService, FacilitiesService, AlertService, AccessControlService

Adapter

Target	AccessControlService
Adapter	AccessControlAdapter
Adaptee	LegacyAccessControlSystem
Client	EmergencyDesk, RestrictAccessCommand, RestoreAccessCommand

Façade

Façade	EmergencyDesk
Subsystem classes	Coordinator, OperatorConsole, SecurityService, MedicalService, FacilitiesService, AccessControlService, AlertService

State – Incident

Context	Incident
State	IncidentStatus
ConcreteState	Reported, Active, Resolved, Cancelled

State – ResponseUnit

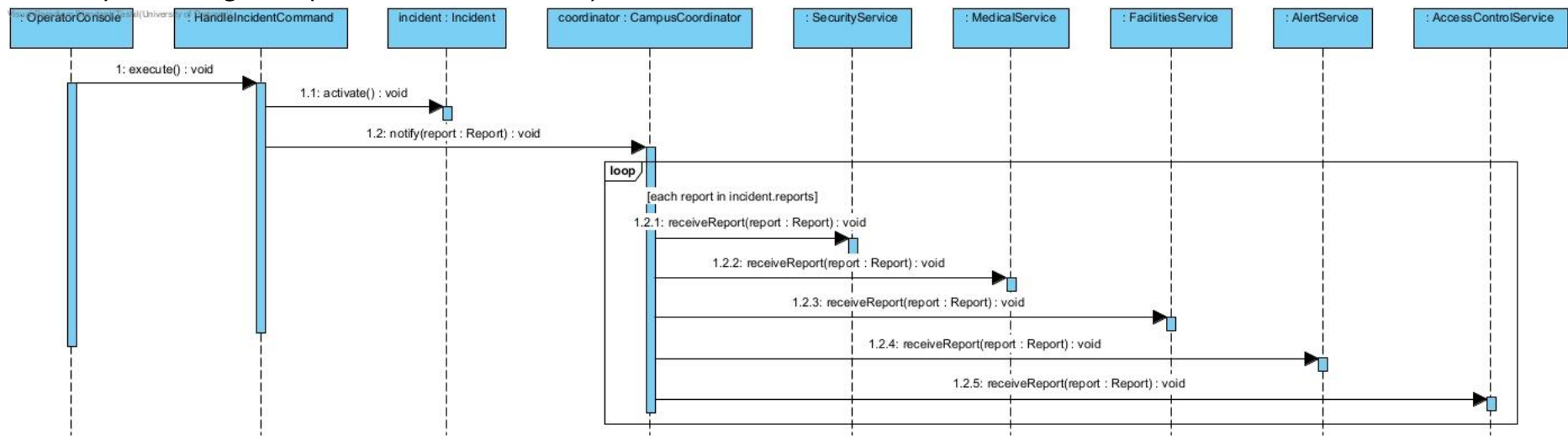
Context	ResponseUnit
State	UnitStatus
ConcreteState	Available, Dispatched, Operating, Unavailable

Composite

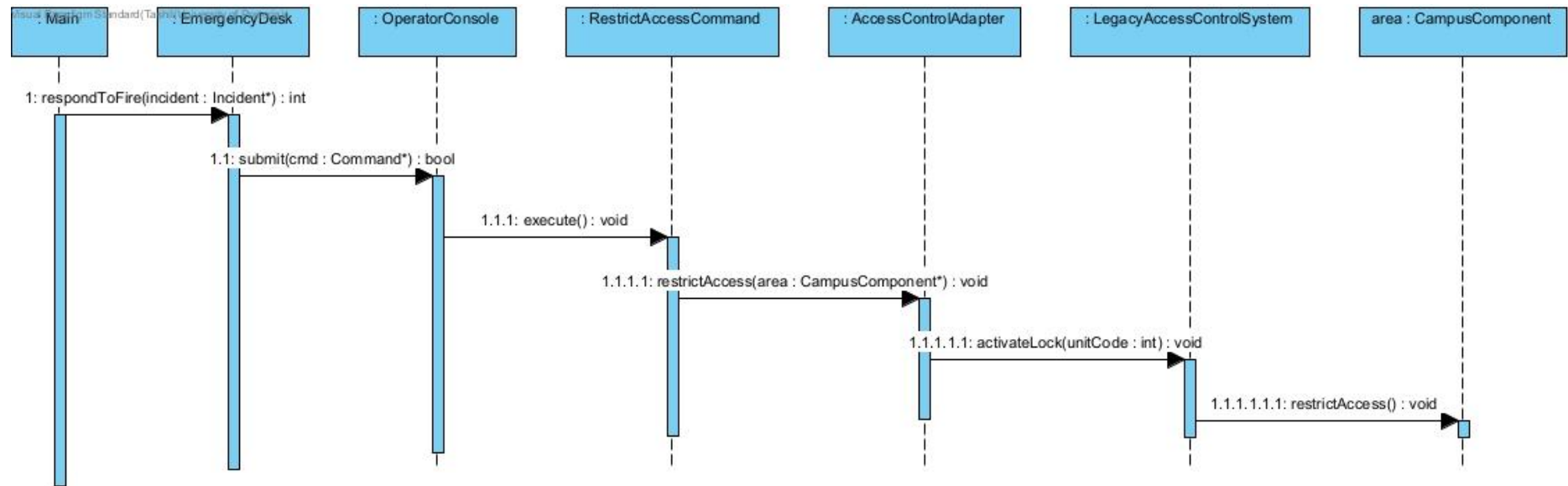
Component	CampusComponent
Leaf	CampusUnit
Composite	CampusZone
Client	LegacyAccessControlSystem, main()

Visual Pedagogy Standard (Tashir/University of Pretoria)

UML Sequence Diagram 1 (Command and Mediator)

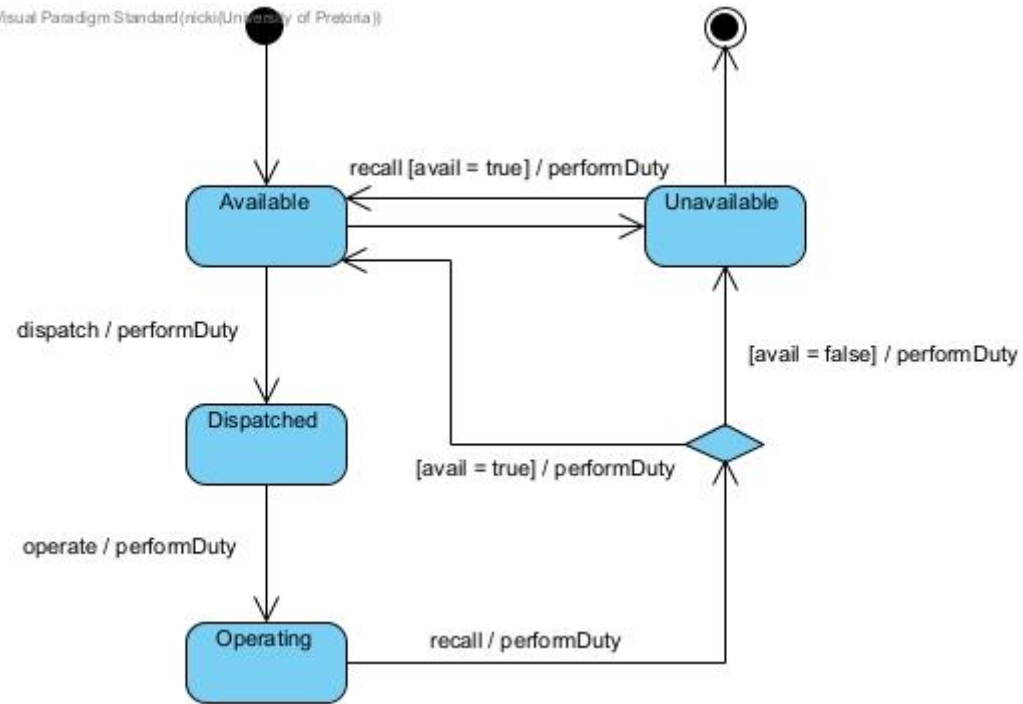


UML Sequence Diagram 2 (Façade and Adapter)



UML State Diagram (ResponseUnit)

Visual Paradigm Standard (ricki@University of Pretoria)



Docker and Valgrind Evidence

```
User@ETERNITY MINGW64 ~/Downloads/UP/COS/214/Pracs/Prac5/COS-214-Prac-5 (dev)
$ docker compose run campusguard make valgrind
Container cos-214-prac-5-campusguard-run-fb818d0ea212 Creating
Container cos-214-prac-5-campusguard-run-fb818d0ea212 Created
valgrind --leak-check=full --keep-stacktraces=alloc-and-free --track-origins=yes ./campusguard
==7== Memcheck, a memory error detector
==7== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==7== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==7== Command: ./campusguard
==7==
```

Running Incident scenario no. 1 (Respond to fire)

=====FIRE EMERGENCY REPORTED=====

[EmergencyDesk] Location: Res Hall A

[EmergencyDesk] Description:

A student had a heart attack at resHallA

Incident is now active.

Security Response Unit 'IBS Security' has been dispatched to Res Hall A.

Response Unit successfully dispatched.

[Console] OK: Dispatch unit to Res Hall A (incident #1000) [IBS Security]

Medical Response Unit 'ABS Medical services' has been dispatched to Res Hall A.

Response Unit successfully dispatched.

[Console] OK: Dispatch unit to Res Hall A (incident #1000) [ABS Medical services]

Facilities Response Unit 'TukTuk central' has been dispatched to Res Hall A.

Response Unit successfully dispatched.

[Console] OK: Dispatch unit to Res Hall A (incident #1000) [TukTuk central]

Access to Campus Component 'Res Hall A' (Code: 12) has been restricted.

[Console] OK: Restrict access to Res Hall A

=====

SENDING ALERT...

Message:

Fire at Res Hall A - avoid the area

=====

[Console] OK: Send alert: Fire at Res Hall A - avoid the area

[EmergencyDesk] Reporting on dispatched units...

Security Response Unit 'IBS Security' is responding to emergency.

Medical Response Unit 'ABS Medical services' is responding to emergency.

Facilities Response Unit 'TukTuk central' is responding to emergency.

Security Response Unit 'IBS Security' is currently on standby.

Medical Response Unit 'ABS Medical services' is currently on standby.

Facilities Response Unit 'TukTuk central' is currently on standby.

Incident has now been resolved.

[EmergencyDesk] Fire response complete: 5/5 steps succeeded

=====

[Main] Operator decides to stand down the last two actions taken...

[EmergencyDesk] Standing down last 2 action(s)

=====

SENDING ALERT...

Message:

ALL CLEAR: Fire at Res Hall A - avoid the area

=====

[Console] UNDO: Send alert: Fire at Res Hall A - avoid the area

Access to Campus Component 'Res Hall A' (Code: 12) has been restored.

[Console] UNDO: Restrict access to Res Hall A

Running Incident scenario no. 2 (HandleIncident)

=====NEW INCIDENT REPORTED=====

[EmergencyDesk] Location: Chemistry Lab

[EmergencyDesk] Description:

Satan's dogs got loose again and are trashing the chem lab

[EmergencyDesk] Coordinating services...

Incident is now active.

Attempting security response unit dispatch...

Security Response Unit 'IBS Security' has been dispatched to Chemistry Lab.

Response Unit successfully dispatched.

Attempting security response unit dispatch...

Response Unit dispatch failed: no available Response Units.

[Console] OK: Handle incident #1001

[EmergencyDesk] Reporting on dispatched units...

Security Response Unit 'IBS Security' is responding to emergency.

Security Response Unit 'IBS Security' is currently on standby.

Incident has now been resolved.

=====

Running Incident scenario no. 3 (Direct Coordinator notify + mid-incident unit reports)

[EmergencyDesk] New incident being managed...

[EmergencyDesk] Location: Gymnasium

[EmergencyDesk] Description:

A gas leak has been detected in the Gymnasium

[EmergencyDesk] Coordinating services...

Incident is now active.

Attempting security response unit dispatch...

Security Response Unit 'IBS Security' has been dispatched to Gymnasium.

Response Unit successfully dispatched.

Attempting facilities response unit dispatch...

Facilities Response Unit 'TukTuk central' has been dispatched to Gymnasium.

Response Unit successfully dispatched.

[Console] OK: Handle incident #1002

[EmergencyDesk] Reporting on dispatched units...

Security Response Unit 'IBS Security' is responding to emergency.

Facilities Response Unit 'TukTuk central' is responding to emergency.

[Main] Dispatched units reporting changes mid-incident...

Attempting facilities response unit dispatch...

Response Unit dispatch failed: no available Response Units.

=====

SENDING ALERT...

Message:

Area Gymnasium unsafe.

=====

Access to Campus Component 'Gymnasium' (Code: 15) has been restricted.

Attempting facilities response unit dispatch...

Response Unit dispatch failed: no available Response Units.

=====

SENDING ALERT...

Message:

Evacuate area Gymnasium .

=====

Access to Campus Component 'Gymnasium' (Code: 15) has been restricted.

[EmergencyDesk] Resolving current incident...

Security Response Unit 'IBS Security' is currently on standby.

Facilities Response Unit 'TukTuk central' is currently on standby.

Incident has now been resolved.

==7==

==7== HEAP SUMMARY:

==7== in use at exit: 0 bytes in 0 blocks

==7== total heap usage: 236 allocs, 236 frees, 82,969 bytes allocated

==7==

==7== All heap blocks were freed -- no leaks are possible

==7==

==7== For lists of detected and suppressed errors, rerun with: -s

==7== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)

GDB Investigation

User@ETERNITY MINGW64 ~/Downloads/UP/COS/214/Pracs/Prac5/COS-214-Prac-5 (dev)

```
$ gdb -q ./campusguard
```

```
Reading symbols from ./campusguard...
```

```
(gdb) break
```

```
No default breakpoint address now.
```

```
(gdb) break EmergencyDesk::standDown
```

```
Breakpoint 1 at 0x140003cc7: file EmergencyDesk.cpp, line 106.
```

```
(gdb) break OperatorConsole::undoLast
```

```
Breakpoint 2 at 0x140007be9: file OperatorConsole.cpp, line 34.
```

```
(gdb) break sendAlertCommand::undo
```

```
Function "sendAlertCommand::undo" not defined.
```

```
Make breakpoint pending on future shared library load? (y or [n]) n
```

```
(gdb) break SendAlertCommand::undo
```

```
Breakpoint 3 at 0x140009243: file SendAlertCommand.cpp, line 14.
```

```
(gdb) break RestrictAccessCommand::undo
```

```
Breakpoint 4 at 0x140008de8: file RestrictAccessCommand.cpp, line 16.
```

```
(gdb) run
```

```
Starting program: C:\Users\User\Downloads\UP\COS\214\Pracs\Prac5\COS-214-Prac-5\campusguard.exe
```

```
[New Thread 30104.0x413c]
```

```
-----  
Running Incident scenario no. 1 (Respond to fire)  
-----
```

```
=====FIRE EMERGENCY REPORTED=====
```

```
[EmergencyDesk] Location: Res Hall A
```

```
[EmergencyDesk] Description:
```

```
A student had a heart attack at resHallA
```

```
Incident is now active.
```

```
Security Response Unit 'IBS Security' has been dispatched to Res Hall A.
```

```
Response Unit successfully dispatched.
```

```
[Console] OK: Dispatch unit to Res Hall A (incident #1000) [IBS Security]
```

```
Medical Response Unit 'ABS Medical services' has been dispatched to Res Hall A.
```

```
Response Unit successfully dispatched.
```

```
[Console] OK: Dispatch unit to Res Hall A (incident #1000) [ABS Medical services]
```

```
Facilities Response Unit 'TukTuk central' has been dispatched to Res Hall A.
```

```
Response Unit successfully dispatched.
```

```
[Console] OK: Dispatch unit to Res Hall A (incident #1000) [TukTuk central]
```

```
Access to Campus Component 'Res Hall A' (Code: 12) has been restricted.
```

```
[Console] OK: Restrict access to Res Hall A
```

```
=====
```

```
SENDING ALERT...
```

```
Message:
```

```
Fire at Res Hall A - avoid the area
```

```
=====
```

```
[Console] OK: Send alert: Fire at Res Hall A - avoid the area
```

[EmergencyDesk] Reporting on dispatched units...
Security Response Unit 'IBS Security' is responding to emergency.
Medical Response Unit 'ABS Medical services' is responding to emergency.
Facilities Response Unit 'TukTuk central' is responding to emergency.
Security Response Unit 'IBS Security' is currently on standby.
Medical Response Unit 'ABS Medical services' is currently on standby.
Facilities Response Unit 'TukTuk central' is currently on standby.
Incident has now been resolved.

[EmergencyDesk] Fire response complete: 5/5 steps succeeded

=====

[Main] Operator decides to stand down the last two actions taken...

Thread 1 hit Breakpoint 1, EmergencyDesk::standDown (this=0x28c8f5a28a0, steps=2) at
EmergencyDesk.cpp:106

```
106      std::cout << "[EmergencyDesk] Standing down last " << steps << " action(s)\n";
```

(gdb) print steps

\$1 = 2

(gdb) continue

Continuing.

[EmergencyDesk] Standing down last 2 action(s)

Thread 1 hit Breakpoint 2, OperatorConsole::undoLast (this=0x28c8f5a2850) at
OperatorConsole.cpp:34

```
34      if (history.empty()) {
```

(gdb) print history.size()

\$2 = 5

(gdb) continue

Continuing.

Thread 1 hit Breakpoint 3, SendAlertCommand::undo (this=0x28c8f5c26e0) at
SendAlertCommand.cpp:14

```
14      alerts->sendAlert("ALL CLEAR: " + message);
```

(gdb) print history.size()

No symbol "history" in current context.

(gdb) continue

Continuing.

=====

SENDING ALERT...

Message:

ALL CLEAR: Fire at Res Hall A - avoid the area

=====

[Console] UNDO: Send alert: Fire at Res Hall A - avoid the area

Thread 1 hit Breakpoint 2, OperatorConsole::undoLast (this=0x28c8f5a2850) at
OperatorConsole.cpp:34

```
34      if (history.empty()) {
```

```
(gdb) print history.size()
$3 = 4
(gdb) continue
Continuing.
```

```
Thread 1 hit Breakpoint 4, RestrictAccessCommand::undo (this=0x28c8f5c25d0) at
RestrictAccessCommand.cpp:16
16      if (!wasRestricted) {
(gdb) print area->getName()
$4 = "Res Hall A"
(gdb) print wasRestricted
$5 = false
(gdb) continue
Continuing.
Access to Campus Component 'Res Hall A' (Code: 12) has been restored.
[Console] UNDO: Restrict access to Res Hall A
```

```
-----
Running Incident scenario no. 2 (HandleIncident)
-----
```

=====NEW INCIDENT REPORTED=====

```
[EmergencyDesk] Location: Chemistry Lab
[EmergencyDesk] Description:
Satan's dogs got loose again and are trashing the chem lab
```

```
[EmergencyDesk] Coordinating services...
Incident is now active.
Attempting security response unit dispatch...
Security Response Unit 'IBS Security' has been dispatched to Chemistry Lab.
Response Unit successfully dispatched.
Attempting security response unit dispatch...
Response Unit dispatch failed: no available Response Units.
[Console] OK: Handle incident #1001
```

```
[EmergencyDesk] Reporting on dispatched units...
Security Response Unit 'IBS Security' is responding to emergency.
Security Response Unit 'IBS Security' is currently on standby.
```

```
Incident has now been resolved.
```

```
=====
-----
Running Incident scenario no. 3 (Direct Coordinator notify + mid-incident unit reports)
-----
```

[EmergencyDesk] New incident being managed...

[EmergencyDesk] Location: Gymnasium

[EmergencyDesk] Description:

A gas leak has been detected in the Gymnasium

[EmergencyDesk] Coordinating services...

Incident is now active.

Attempting security response unit dispatch...

Security Response Unit 'IBS Security' has been dispatched to Gymnasium.

Response Unit successfully dispatched.

Attempting facilities response unit dispatch...

Facilities Response Unit 'TukTuk central' has been dispatched to Gymnasium.

Response Unit successfully dispatched.

[Console] OK: Handle incident #1002

[EmergencyDesk] Reporting on dispatched units...

Security Response Unit 'IBS Security' is responding to emergency.

Facilities Response Unit 'TukTuk central' is responding to emergency.

[Main] Dispatched units reporting changes mid-incident...

Attempting facilities response unit dispatch...

Response Unit dispatch failed: no available Response Units.

=====

SENDING ALERT...

Message:

Area Gymnasium unsafe.

=====

Access to Campus Component 'Gymnasium' (Code: 15) has been restricted.

Attempting facilities response unit dispatch...

Response Unit dispatch failed: no available Response Units.

=====

SENDING ALERT...

Message:

Evacuate area Gymnasium .

=====

Access to Campus Component 'Gymnasium' (Code: 15) has been restricted.

[EmergencyDesk] Resolving current incident...

Security Response Unit 'IBS Security' is currently on standby.

Facilities Response Unit 'TukTuk central' is currently on standby.

Incident has now been resolved.

[Thread 30104.0x413c exited with code 0]

[Inferior 1 (process 30104) exited normally]

GitHub Repository Link

<https://github.com/KatNikKing/COS-214-Prac-5.git>

Collaboration Reflection

The work was divided according to the six design patterns, with each team member implementing their assigned components in their own Git branch. Simi implemented the Mediator, Adapter and Composite patterns. Tashil implemented the Command and Façade patterns. Simon implemented the State pattern. This allowed the team to work independently while keeping the different parts of the project organised. Branches were merged into dev once each implementation was completed, and the completed project was then pushed to main. All merges went through pull requests rather than being pushed directly, giving the team a chance to review changes before they landed. GitHub issues were used throughout to track what still needed to be done and to tick off work as it was completed. Collaboration also took place through reviewing each other's pull requests and making shared design decisions.

Contribution Statement

All three members contributed substantially. Simi implemented the Mediator, Adapter and Composite patterns, wrote the Makefile, and produced the first sequence diagram and the state diagram. Tashil implemented the Command and Façade patterns, produced the second sequence diagram, and compiled the PDF. Simon implemented the State pattern, carried out testing, configured the Docker build, wrote the README, and ran the Valgrind and GDB investigation. All three collaborated on the overall system design and the UML class diagram.